Arash NasrEsfahani

Department of Mathematics, Statistics, and Computer Science, University of Tehran, Tehran, Iran

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Research Interests

- Human-Computer Interaction
- Machine Learning
- Data Analysis
- Human-Robot Interaction

Education

University of Tehran *Sep 2024 – Present*
MSc in Computer Science (Artificial Intelligence) – GPA: 19.36/20 (4.0/4.0)

University of Tehran *Sep 2021 – Aug 2024*
BSc in Computer Science – GPA: 15.85/20 (3.12/4.0)

University of Tehran *Sep 2019 – Aug 2021*
BSc in Electrical Engineering (Transferred to Computer Science)

Publications

Embodied AI for Kitchen Scene Rearrangement: A Deep Spatio-Temporal Approach 2026
Published in Intelligent Service Robotics
Arash Nasr Esfahani, Hamed Hosseini, Mehdi Tale Masouleh, Ahmad Kalhor

NeXtSwin-X: Dual-Branch Cross-Attention Fusion of ConvNeXt and Swin Transformer for Accurate Brain Tumor Classification from MRI and CT 2026
Accepted for publication in Scientific Reports
Arash Nasr Esfahani, Amirreza Jalili, Hedieh Sajedi

Ensemble of Foundation Models for Sensor-Based Locomotion and Transportation Mode Recognition 2025
Published in UbiComp Companion 25: Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing
Arash Nasr Esfahani, Mohammad Foad Abdi, Yousef Alikhani, Mohammad Mahdi Azizi, Bagher BabaAli, Mehdi Mohhebi

AI-Driven Relocation Tracking in Dynamic Kitchen Environments 2024
Published in ICCKE 2024: 14th International Conference on Computer and Knowledge Engineering
Arash Nasr Esfahani, Hamed Hosseini, Mehdi Tale Masouleh, Ahmad Kalhor, Hedieh Sajedi

Experience

The Human and Robot Interaction Lab (TaarLab) *University of Tehran*
Researcher *Sep 2023 – Present*
Under Supervision of: Prof. M. Tale Masouleh
As part of a research team on scene understanding and manipulation, I led a project on robotic scene rearrangement in dynamic environments using the AI2-THOR simulator, applying computer-vision techniques such as temporal frame analysis and deep-learning object detection to rearrange objects across virtual scenes.

KareOne (Business Intelligence Company) *Tehran, Iran*
AI Engineer *Sep 2025 – Present*
Helped design the company's overall AI strategy and adoption, and developed its main RAG chatbot, which lets a user query a large collection of their own documents and get answers grounded in them. Also built data-analysis tools for finding competitors and turning numeric and textual data into business insights and financial-data visualizations, all centered on explainable AI and clear, interpretable insights.

Department of Mathematics, Statistics, and Computer Science *University of Tehran*
Teaching Assistant
- **Digital Image Processing** *(Graduate)* — Prof. H. Sajedi, Spring 2026: designed homework assignments and oversaw students' course projects, providing guidance and feedback throughout.

- **Database Management Systems** — Prof. M. Goodarzi, Fall 2023: developed course materials including homework and projects, and supervised a group of students in their final projects on database design.

Department of Mathematics, Statistics, and Computer Science *University of Tehran*
Researcher *April 2025 – July 2025*
Under supervision of: Prof. B. Babaali
Collaborated in a five-person team to compete in the SHL 2025 Human Activity Recognition Challenge, hosted by the University of Sussex. Our team achieved **1st place**, with results presented at the HASCA Workshop, Ubicomp 2025. My primary role involved applying time series foundation models and advanced data analysis techniques to improve model performance.

Honors and Awards

- Ranked 598 among more than 160000 participants in the Nationwide Iranian University Entrance Exam in Mathematics and Physics
- Ranked 96 among more than 2000 participants in the Nationwide Entrance Exam for a Master's Degree in Computer Science
- Advanced to the second round of the Iranian Student Olympiads in Computer Science, Physics, and Persian Literature, an achievement reached by less than 10% of initial participants.

Notable Courses

• **Machine Learning** ***(Graduate)*** 19.41/20
Prof. B. Babaali - Fall 2024

• **Advanced Algorithms** ***(Graduate)*** 19.51/20
Prof. M. Mohammad-Noori - Fall 2024

• **Artificial Intelligence** 18.6/20
Prof. H. Sajedi - Fall 2022

• **Graph Theory and Applications** 19/20
Prof. M. Mohammad-Noori - Fall 2022

• **Digital Image Processing** ***(Graduate)*** 20/20
Prof. H. Sajedi - Spring 2023

• **Natural Language Processing** ***(Graduate)*** 19.4/20
Prof. B. Babaali - Spring 2024

Selected Projects

Deep Learning Framework for Human Activity Recognition | Python, TensorFlow, Keras

Implemented a hybrid CNN–LSTM model with self-attention for wearable sensor-based activity recognition, achieving 95.24% accuracy on UCI-HAR and 98.24% on MHEALTH datasets.

Object Segmentation with U-Net & Seg-Grad-CAM | Python, U-Net, Grad-CAM

Developed a U-Net model for semantic segmentation of horses using the Weizmann Horse Dataset, tested multiple loss functions (Dice, BCE, Focal), achieved 86.11% validation IoU, and visualized model focus areas using Seg-Grad-CAM.

Persian Sentiment Analysis with BERT Variants | Python, Transformers, PyTorch

Fine-tuned ParsBERT and multilingual BERT for sentiment classification on Persian Snappfood reviews, evaluated cross-domain performance on Digikala reviews, with ParsBERT outperforming in all metrics.

Technical Skills

Programming Languages: Python, C/C++, Java, SQL, R, Assembly, HTML/CSS, Verilog, LaTeX

Technologies/Frameworks: Git, Docker, Linux, Tensorflow, Transformers, Sci-kit, PyTorch, Keras, NumPy, Pandas, Matplotlib, Seaborn

Languages

English: Full Professional Proficiency — **TOEFL**: 109 (R:30, L:30, S:25, W:24)

Persian: Native **German:** Elementary Proficiency